

3.1 Introduction

Project Name	Maximum Power Extraction from a wind energy system using Artificial Neural Network (ANN)
University	Ahsanullah University of Science and Technology
Department	Department of Electrical and Electronic Engineering
Location	Dhaka, Bangladesh
Duration	1 Year
Position	Team Leader

3.2 Background

A. Overview

Maximum Power Point Tracking (MPPT) is a process that improves the power obtained from a Wind Energy Conversion System (WECS) through changing the operating point of the system in response to changes in wind conditions. Artificial neural networks (ANN) are used to approximate dynamic relationships between wind velocity, generated turbine power, and conditions, which assist in maximizing extracted power. Proper prediction of wind speed is essential in the production of a renewable energy system. The power extraction efficiency will be enhanced when powerful algorithms such as Genetic Algorithm (GA) and adaptive control systems are utilized. The design of the controller provides the system with real-time functioning; the neural network is fine-tuned with the help of backpropagation algorithms. Forecasting models and nonlinear modeling also contribute to the adaptivity and fault resistance of the system, guaranteeing the optimal performance in different circumstances.

The project aimed at enhancing the maximum power extraction through an ANN-based MPPT based on a wind energy system. At first, wind energy systems were investigated, and an ANN-based MPPT was presented as an adequate approach toward effective power harvesting. After that, the principles of wind

energy conversion, types of turbines, and the impact on the environment were studied in order to gain a better understanding of system performance. Thereafter, the artificial neural networks were examined, and wind turbine data were set up to be properly modeled and trained by an ANN. Then, ANN-based MPPT was developed and coded in MATLAB to determine wind speed and optimum rotor speed. Lastly, the system performance was determined, the errors were reviewed, and maximum power tracking with good performance was confirmed under a different wind condition.

B. Objective

The key aims of the project were to examine wind energy, analyze turbine systems, create an ANN-based MPPT approach, calculate wind and rotor speed in a precise manner, enhance the maximum power extraction under varying wind conditions, and analyze the performance of the system. Its secondary objectives were: -

- To minimize the application of mechanical sensors and decrease system cost.
- To achieve system reliability and accuracy during varying wind conditions.
- To promote the use of clean energy and minimize its effect on the environment.

C. Nature of Work

In this project, I was involved in the research and comprehension of wind energy systems and their derivation of the highest power output. I initially collected data and studied wind energy conversion, various types of turbines, and the environment to create a clear understanding of the system. Afterwards, I learned about artificial neural networks and prepared turbine data to be modeled and trained properly. Subsequently, I developed and coded an ANN-based MPPT in MATLAB to approximate the wind speed and estimate the optimal rotor speed. Lastly, I tested the performance of the system, examined the errors, and ensured that the system was able to trace maximum power under varying wind conditions.

D. Hierarchical Structure

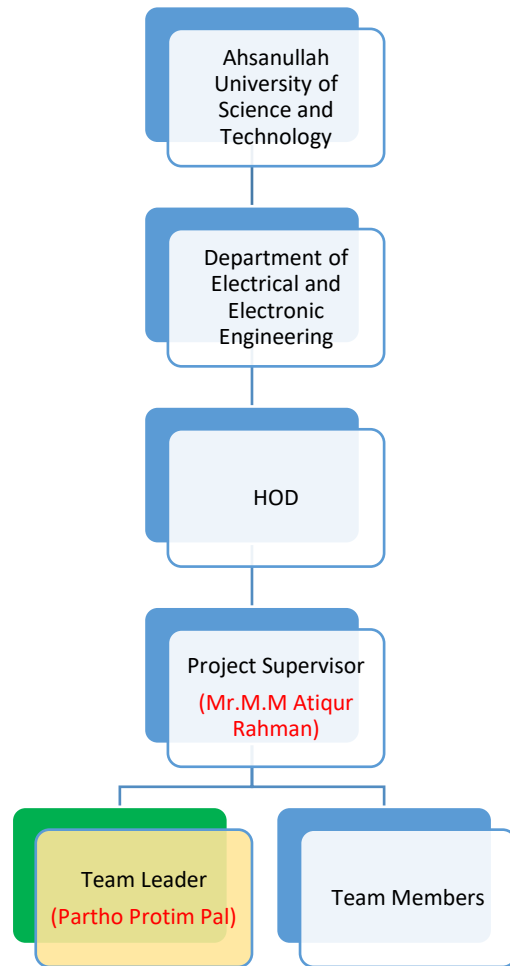


Figure 1: Organizational Hierarchy

E. Duties and Responsibilities

- To analyze wind energy systems and to determine an ANN-based MPPT to achieve maximum power extraction.
- To evaluate wind energy conversion, types of turbines, and environmental influences to understand the system more effectively.
- To examine artificial neural networks and to prepare wind turbine data with correct ANN-based modeling and training.
- To develop and apply an ANN-based MPPT using MATLAB to estimate the wind speed and optimal rotor speed.
- To assess the work of the system, examine the errors, and ensure good maximum power tracking in different wind conditions.

3.3 Personal Engineering Activities (PEA)

A.

I began this work with the investigation of wind energy and the primary issue of peak power output of a wind energy changing system. I read about solar heating as a source of wind energy production, turning wind energy into mechanical energy using wind turbines, and then turning mechanical energy into electrical energy with generators. I described the key components of a wind turbine, including the tower, blades, gearbox, nacelle, controller, and brake, to be able to comprehend the functioning of the system as a whole. I also learned the positive and negative aspects of wind energy, such as its renewability, environmental friendliness, erratic wind speed, and costly installation. I then looked at the MPPT methods such as TSR, OT, PSF, HCS, INC, fuzzy logic, and ANN-based control and compared their advantages and disadvantages. Based on this research, I realized that ANN-based MPPT was appropriate since it was able to approximate the wind velocity and rotor velocity without relying on sensors entirely. In this section, I demonstrated my understanding of engineering basics, mathematics, design concepts, modern technology advancement, and sustainable practice.



Figure 2: Wind Turbines

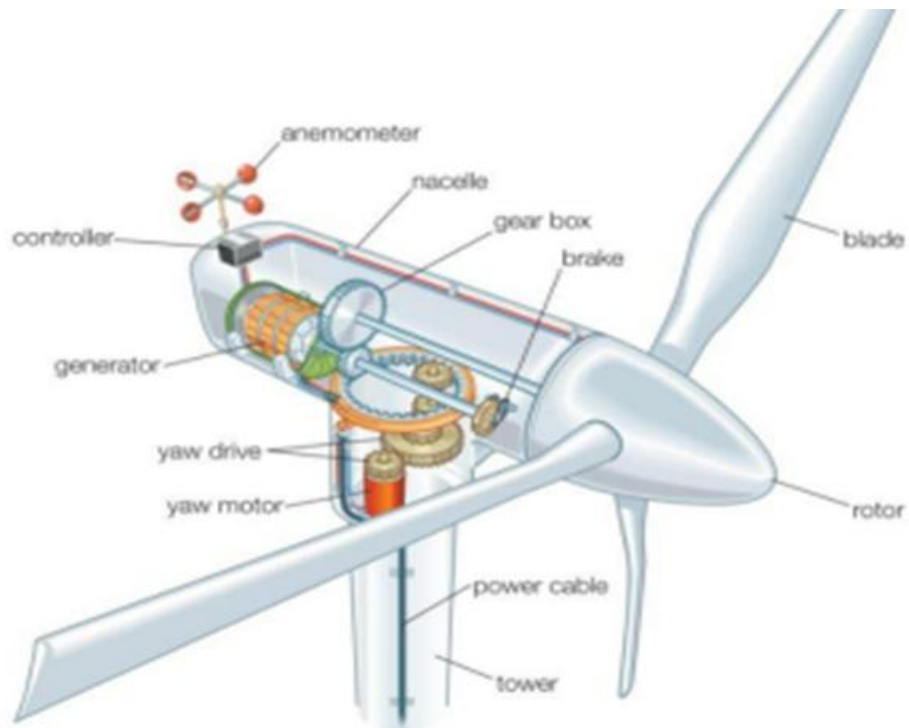


Figure 3: Components of a wind Turbine.

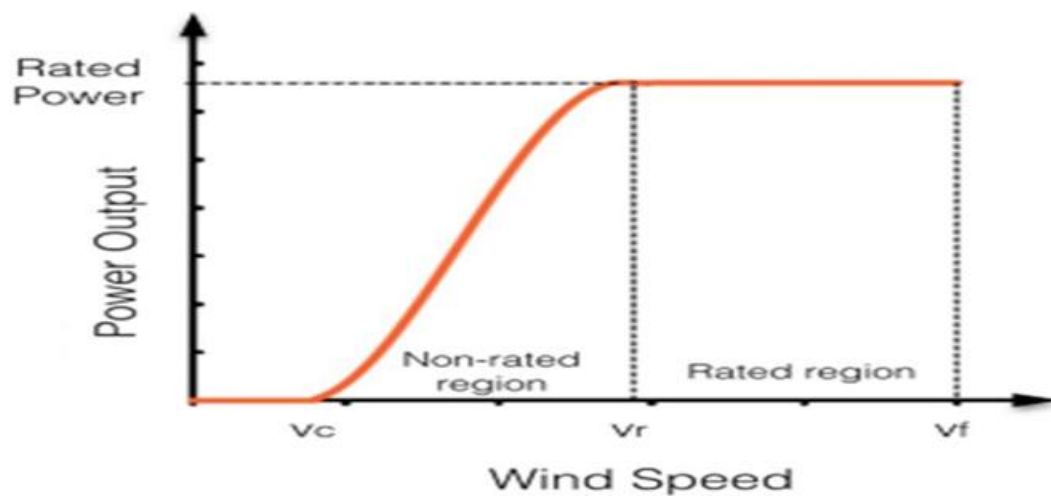


Figure 4: Wind Turbine power curve

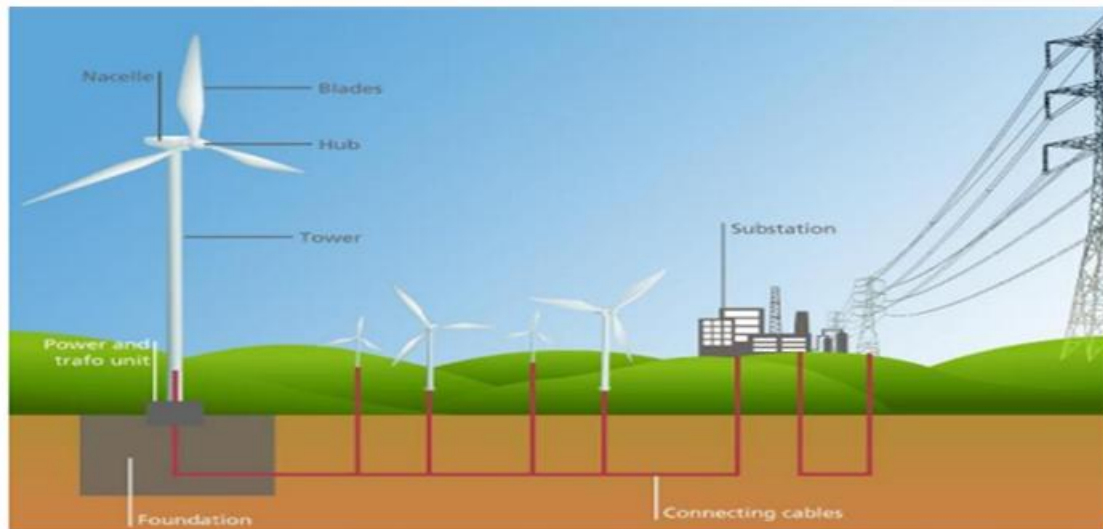


Figure 5: Electricity generation from wind energy

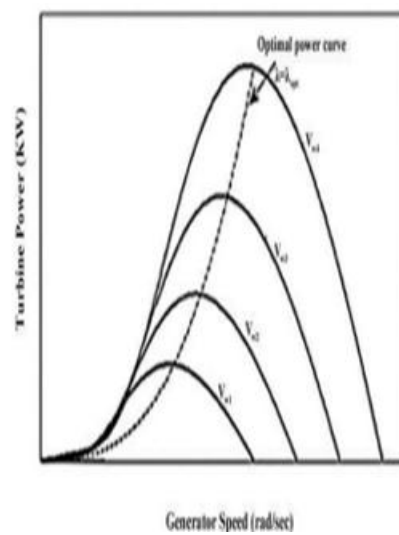


Figure 6: MPPT

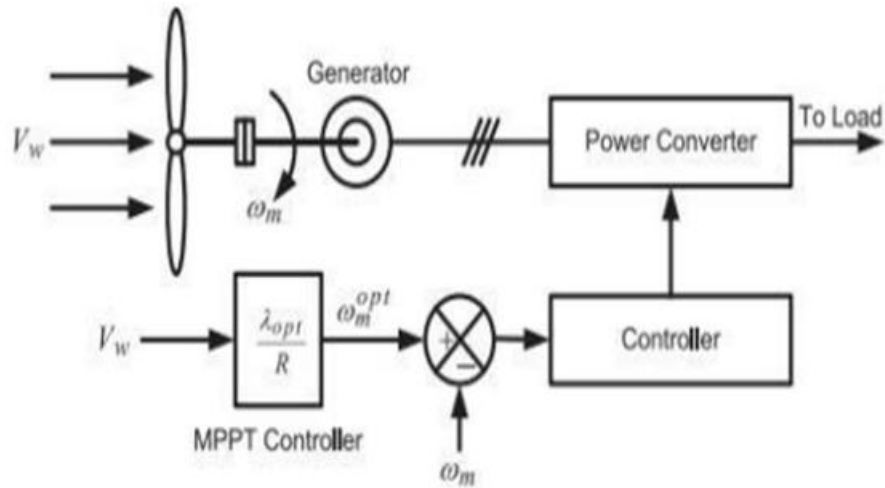


Figure 7: TSR MPPT algorithm of WECS

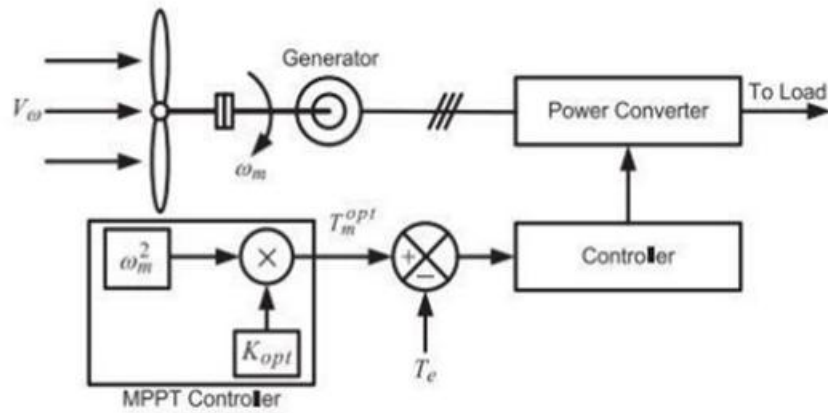


Figure 8: OT MPPT algorithm of WECS

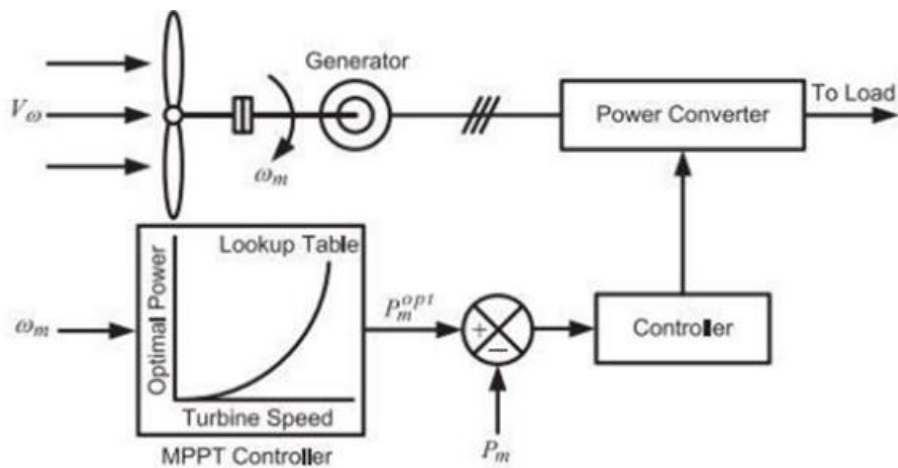


Figure 9: PSF MPPT algorithm of WECS

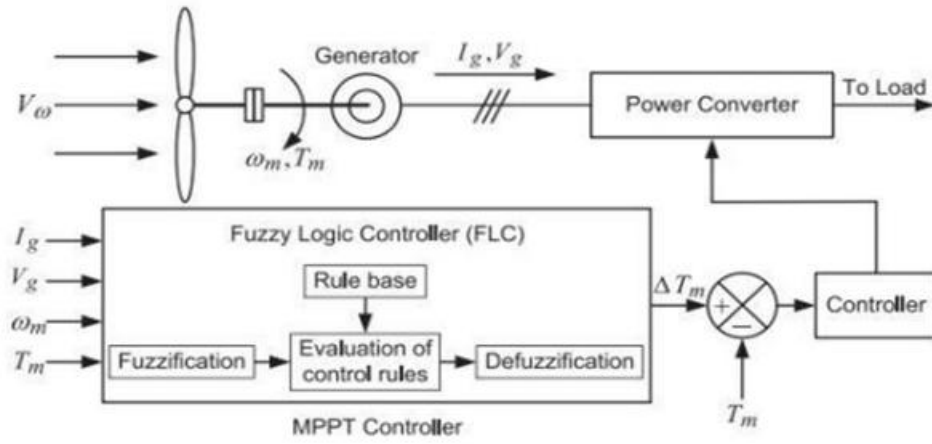


Figure 10: Fuzzy-based MPPT algorithms of WECS

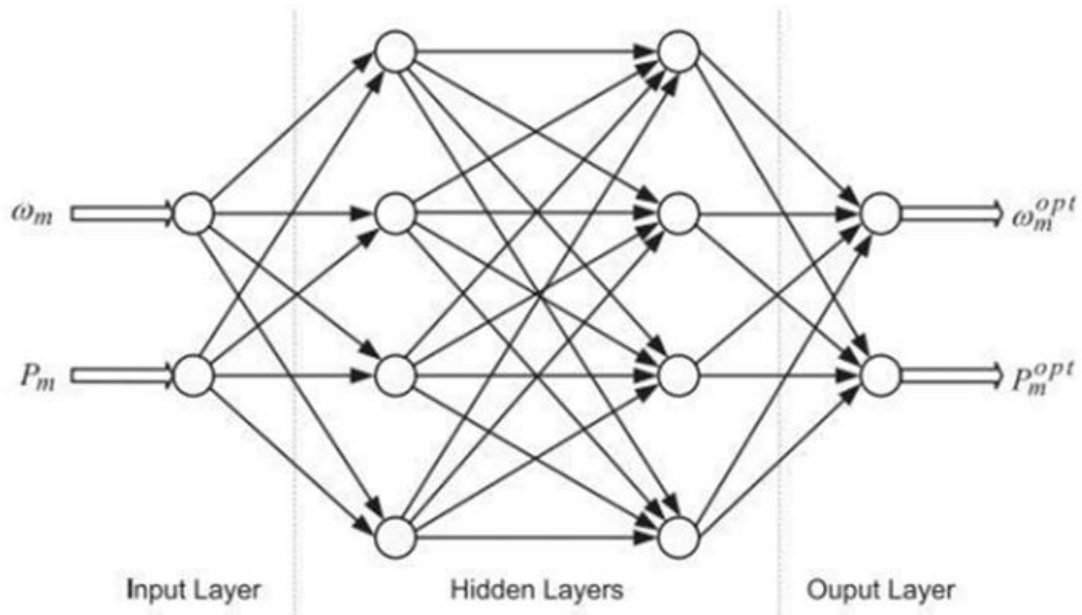


Figure 11: Structure of ANN

Table 1: Diversity of Different MPPT Algorithms

MPPT Algorithms	Sensitivity/Accuracy	Parameters Dependability
ANN(Artificial Neural Network)	High(Up to 98%)	Varies
P&O(Fixed Perturbation size)	Low	On Voltage & Current
P&O(Variable Perturbation size)	High(Up to 96%)	On Voltage & Current
TSR(Tip Speed Ratio)	Low	On Wind Speed
FL(Fuzzy Logic)	High	Varies

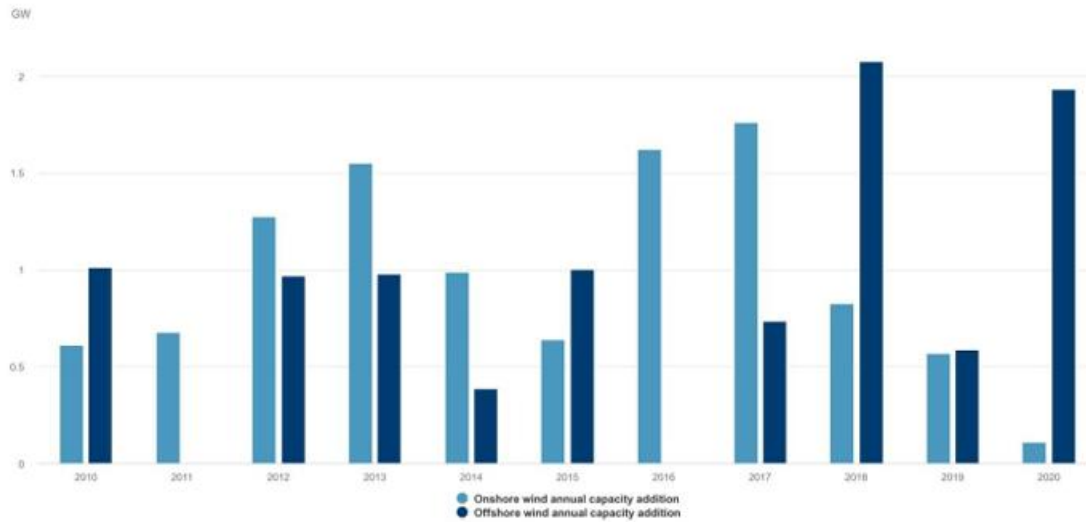


Figure 12: Market status from 2010 to 2020

Total global new wind power installations in 2021

93.6 GW new wind power capacity was added worldwide in 2021, only 1.8% lower than 2020's record year.

This brings total installed wind capacity to **837 GW**, a growth of 12.4% compared to 2020!

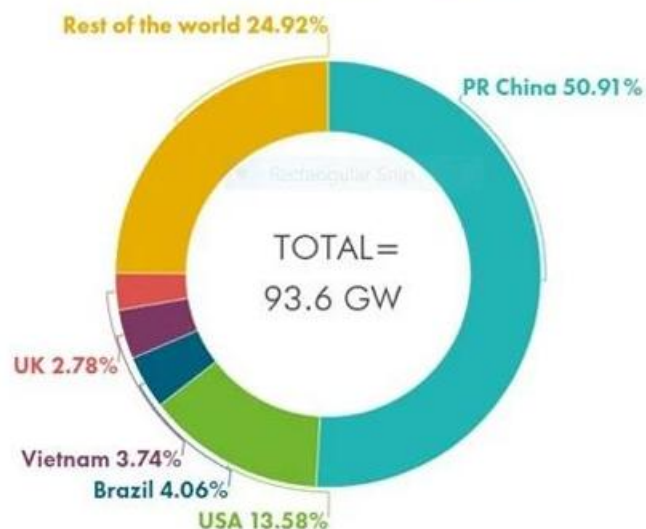
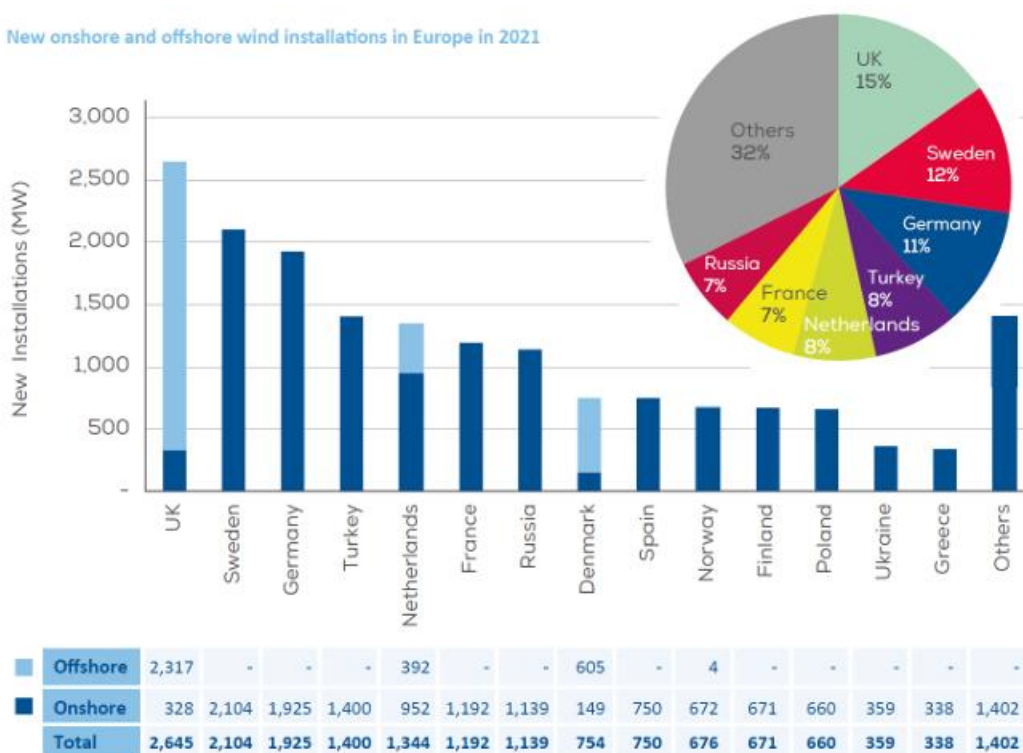


Figure 13: Status of wind power capacity

New onshore and offshore wind installations in Europe in 2021



Source: WindEurope

Figure 14: Comparison of onshore/offshore wind installation in the top countries

Onshore Wind Energy Market, By Region 2020-2027

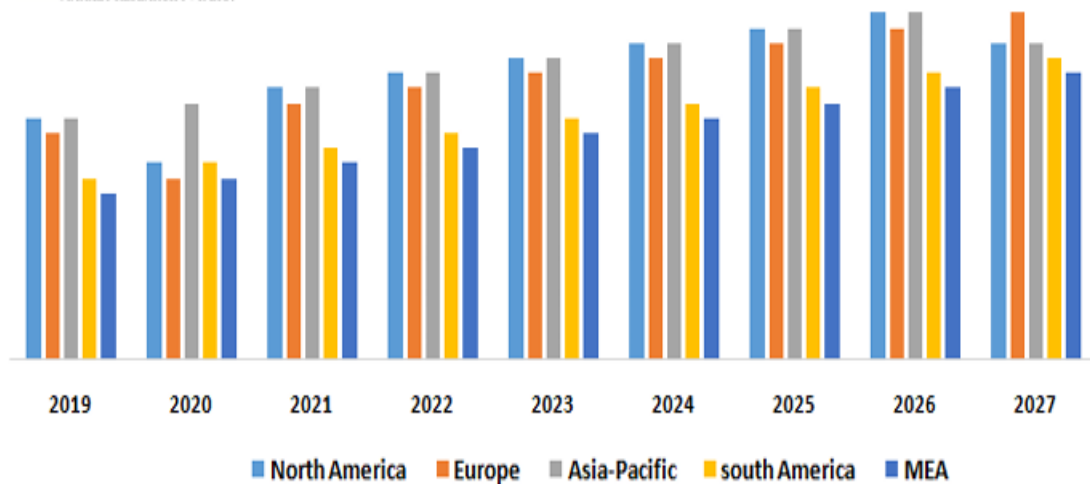


Figure 15: Onshore estimated wind energy market

B.

I then performed document analysis to establish a broad perspective on renewable energy, wind energy conversion, electricity, and turbine development. I learned that the conversion systems of wind energy convert wind energy into mechanical energy and then to electrical energy. I described the fundamental concept of our conversion, the lift and drag forces of blades, and I learned that the force of wind had a very strong impact on any power generation since power was proportional to the cubes of wind speed. I also investigated generators, photovoltaic effect, a turbine system, and environmental impacts so that I might be able to position my project within a larger context of energy. I also studied the historical background of wind turbines, the development of modern wind turbines, and the distinction between the horizontal-axis and vertical-axis turbines, along with their characteristics and drawbacks. This review made me realize that horizontal-axis turbines were often utilized, and layout, place and repairs mattered to a healthy performance. In this work, I applied engineering investigation, interpretation, and evaluation, along with social, environmental, and economic considerations. During the activities, I utilized engineering knowledge, analyzed technical information, and also factored in sustainability and practical design criteria.

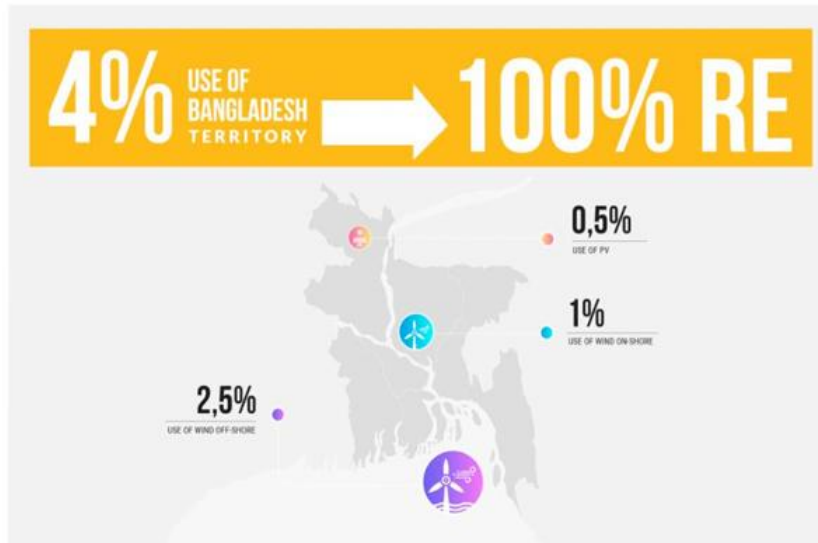


Figure 16: Renewable Energy in Bangladesh

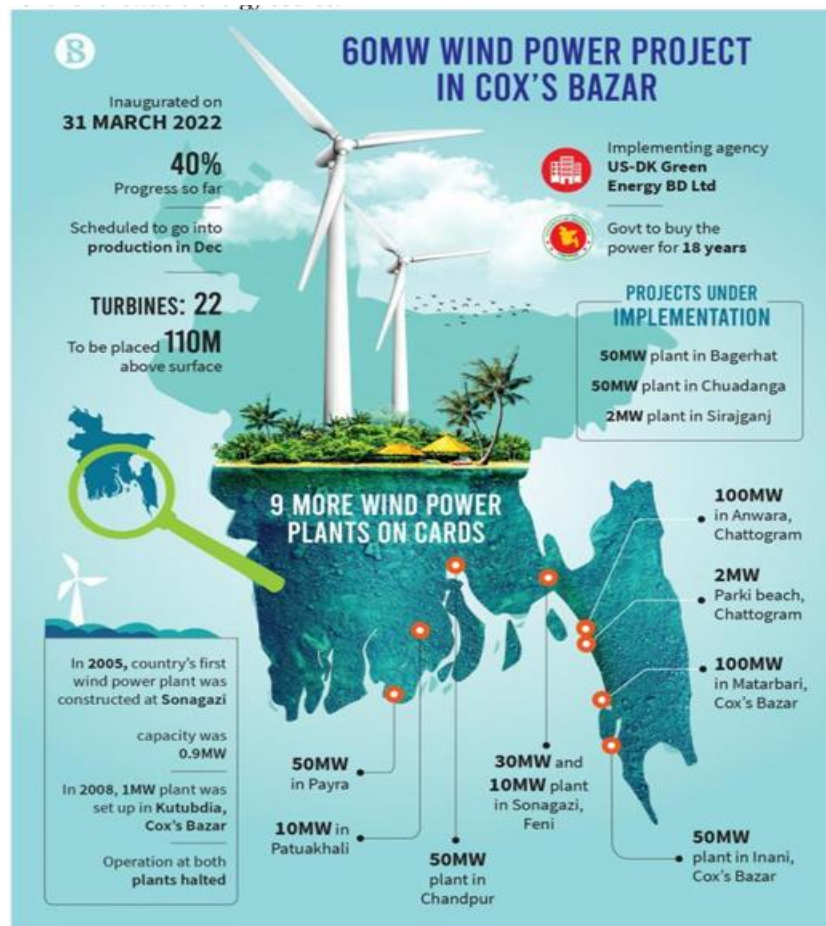


Figure 17: A 60MW Wind Power project

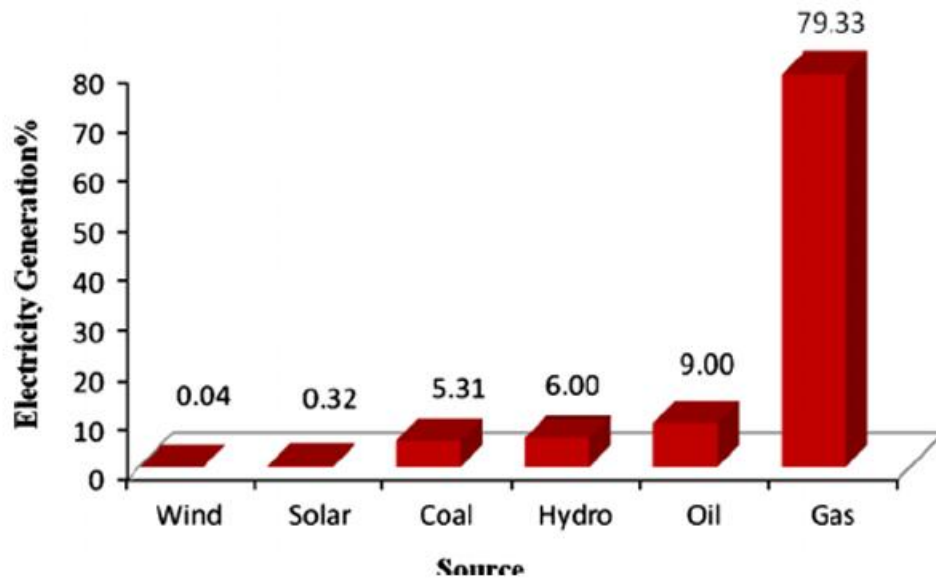


Figure 18: Renewable energy source of Bangladesh

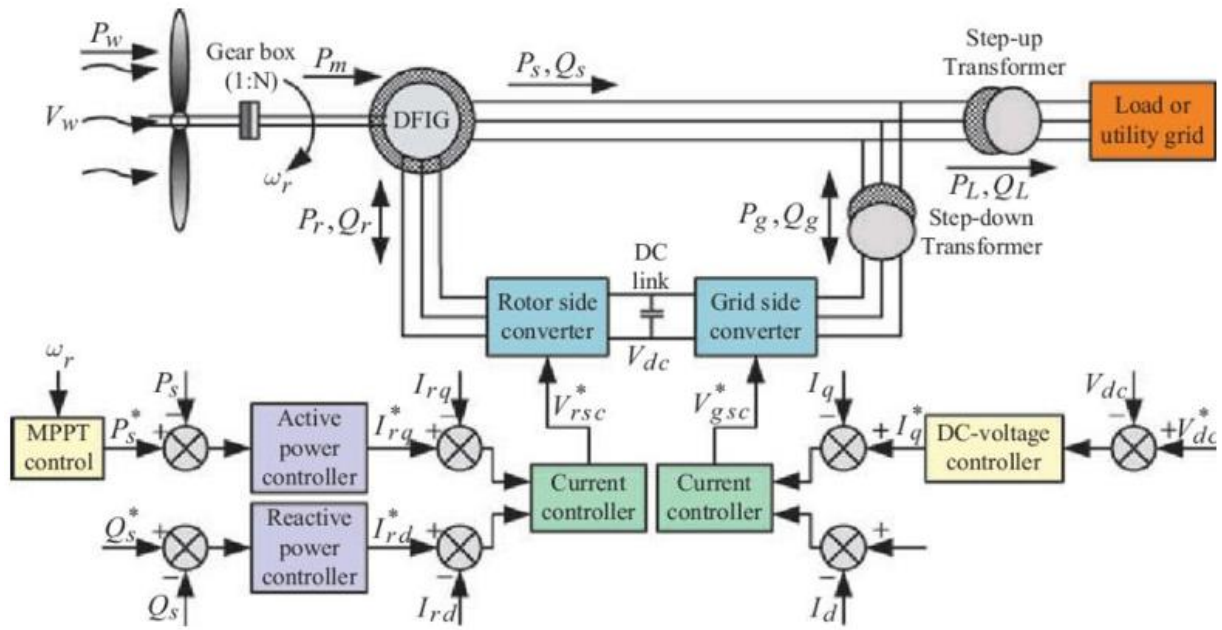


Figure 19: Wind energy conversion principle

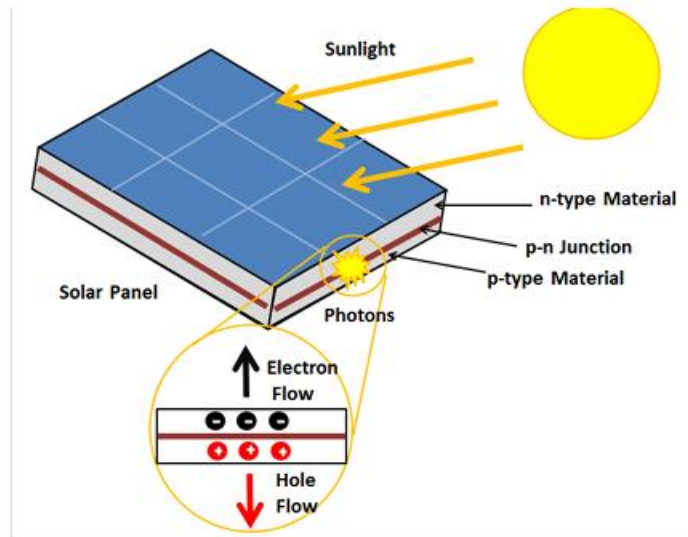


Figure 20: Photovoltaic effect



Figure 21: 1-GW of major China Coal-fired plant



Figure 22: Jebel Ali Power and Desalination Gas power plant



Figure 23: Jaisalmer wind turbine



Figure 24: Three Gorges dam, China



Figure 25:Ruppur nuclear power plant



Figure 26: The Geysers Geothermal field, California

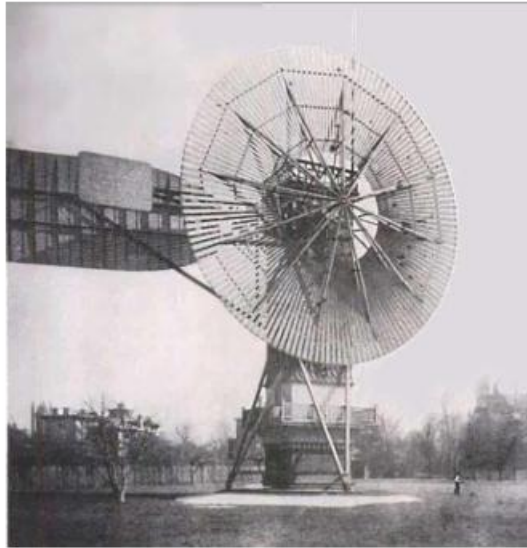


Figure 27: Charles Brush's windmill, Ohio

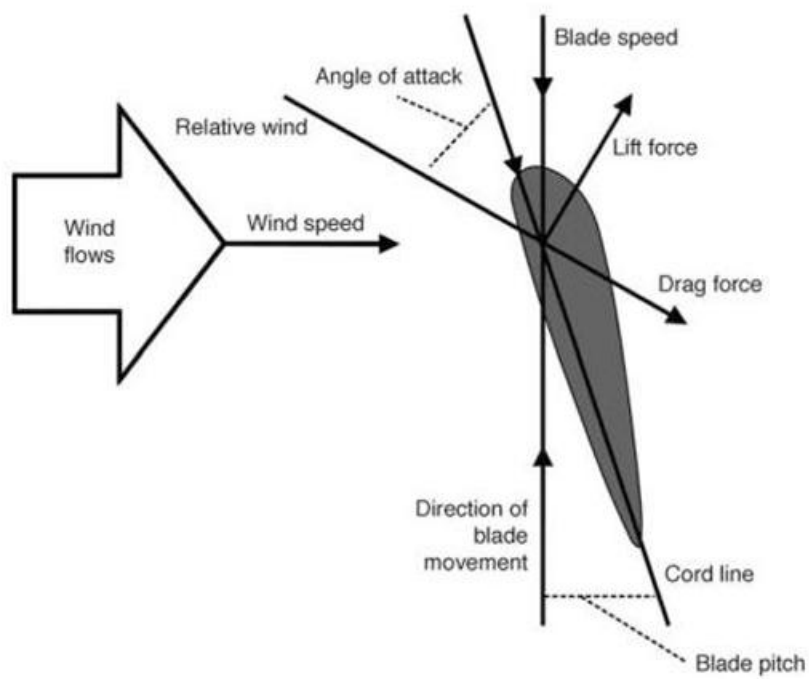


Figure 28: Illustration of the lift and drag force acting on an airfoil blade

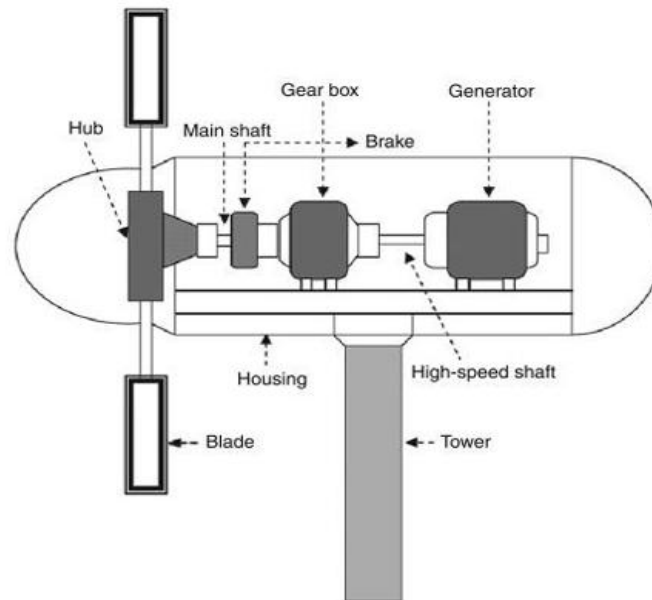


Figure 29: Sectional view of Horizontal Axis Wind Turbine



Figure 30: Vertical Axis Wind Turbine

C.

Once the energy background was built, I researched artificial neural networks thoroughly to ensure that I could use them appropriately in the project. I was initially introduced to the concept of an ANN and its role as a simplified version of the human brain to solve nonlinear problems. I examined the history of its development up to the present machine learning application. Later, I read about the ANN structure, the input layer, the hidden layer, and the output layer, and I understood how weights, bias, activation functions, and backpropagation could assist the network to learn using data. I also examined a few ANN models, which included a multilayer perceptron, a radial basis function network, a Kohonen network, a recurrent neural network, a convolutional neural network, and a modular neural network. Then I considered the sigmoid activation function and the backpropagation algorithm due to their direct relation to my implementation. I also learned the equation of wind turbine power and the relationship between air density, rotor swept area, power coefficient, and wind speed. Then I used MATLAB to extract wind turbine characteristic data, taking a wind speed range of 6-18 m/s with a 0.01 step. This section entailed the application of mathematics, expert knowledge, information handling, and calculating devices carefully.

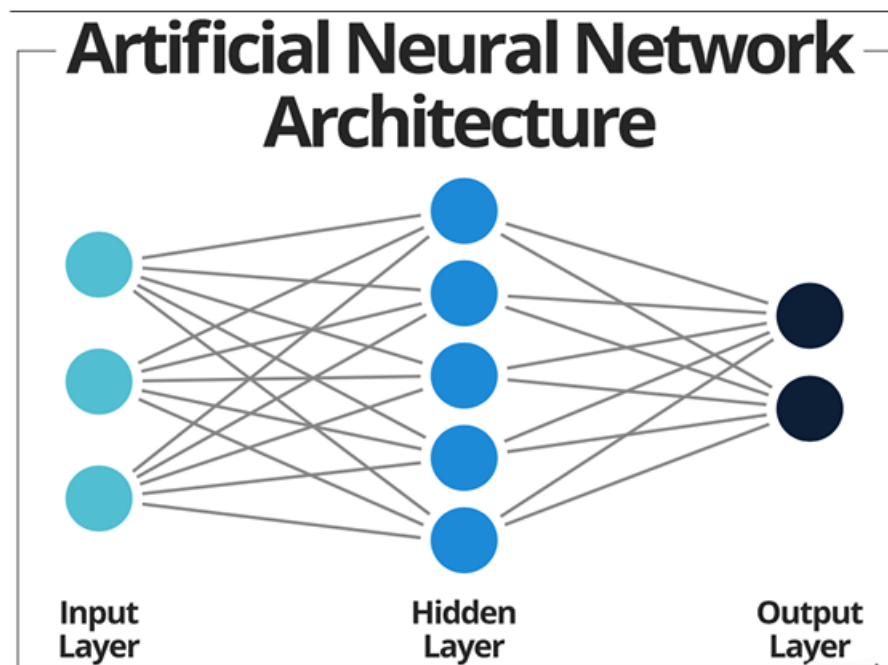


Figure 31: Artificial Neural Networks

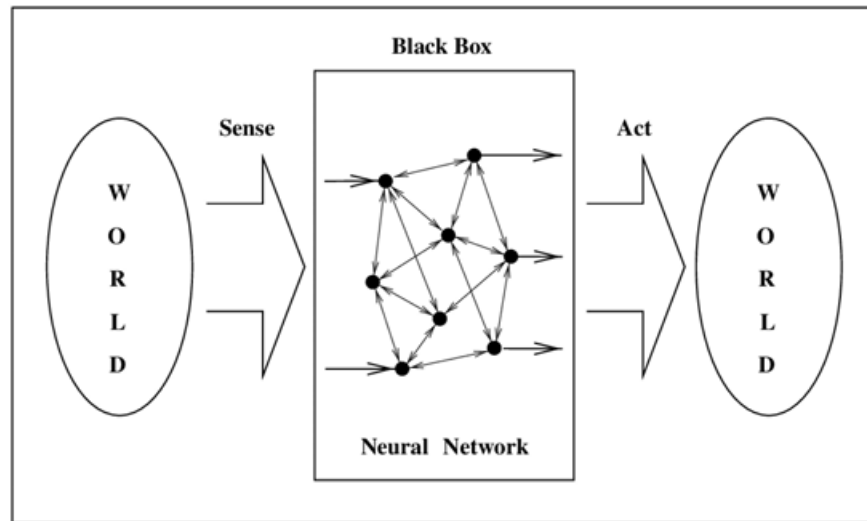


Figure 32: Neural network as a black-box

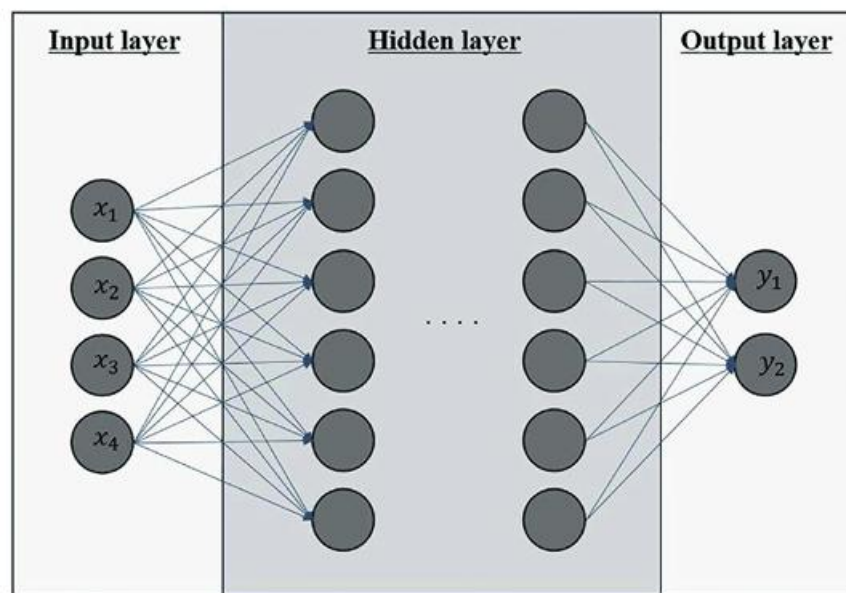


Figure 33: Different Layer of the ANN

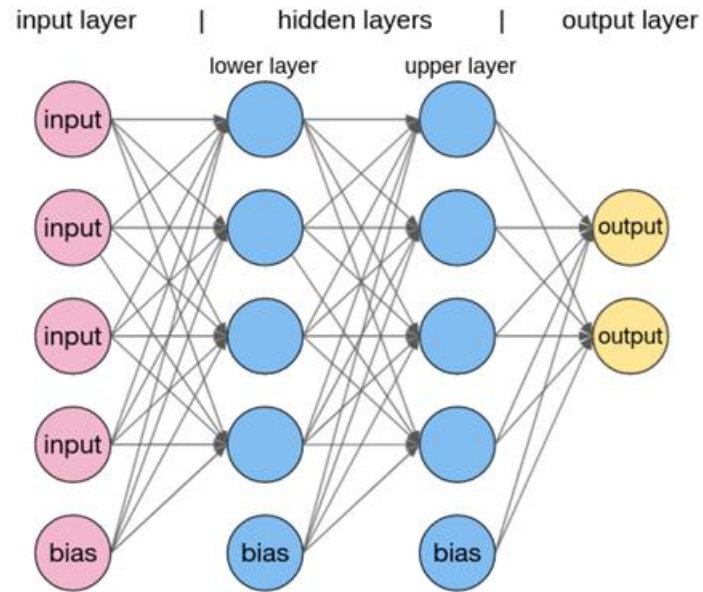


Figure 34: Multilayer Perceptron

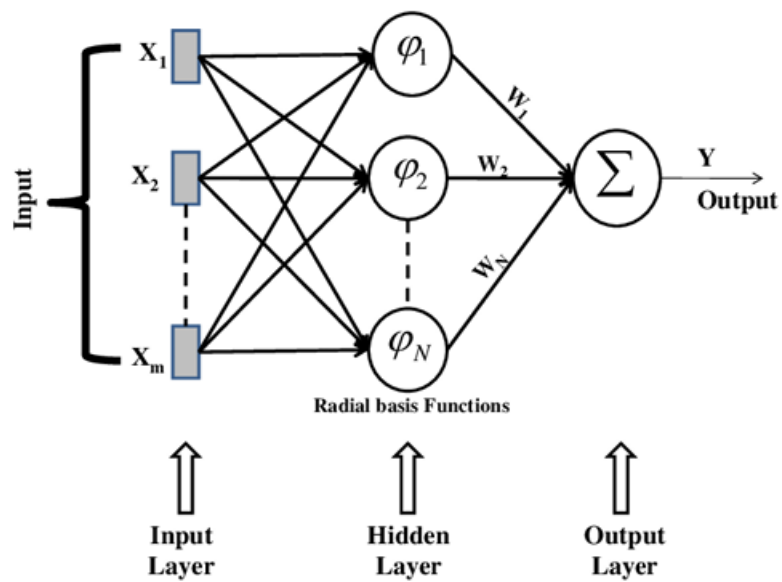


Figure 35: Radial Basis Functions Neural Network

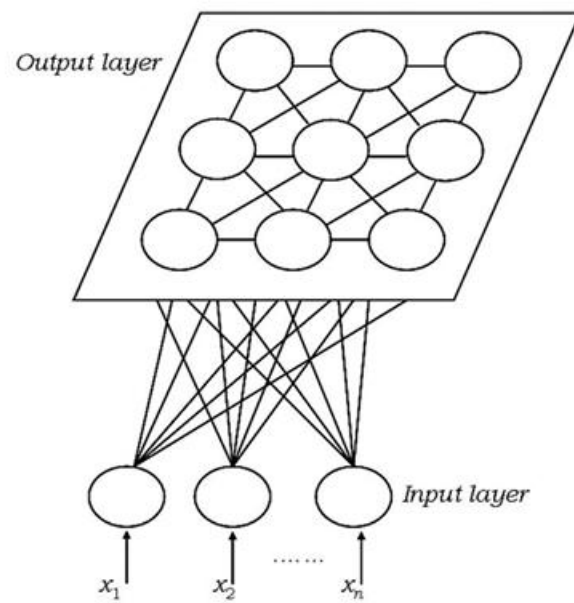


Figure 36: Kohonen Self-Organizing

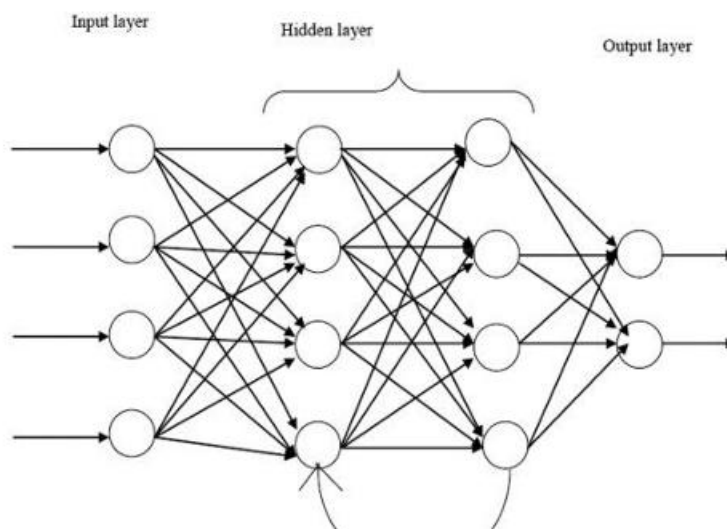


Figure 37: Recurrent Neural Network

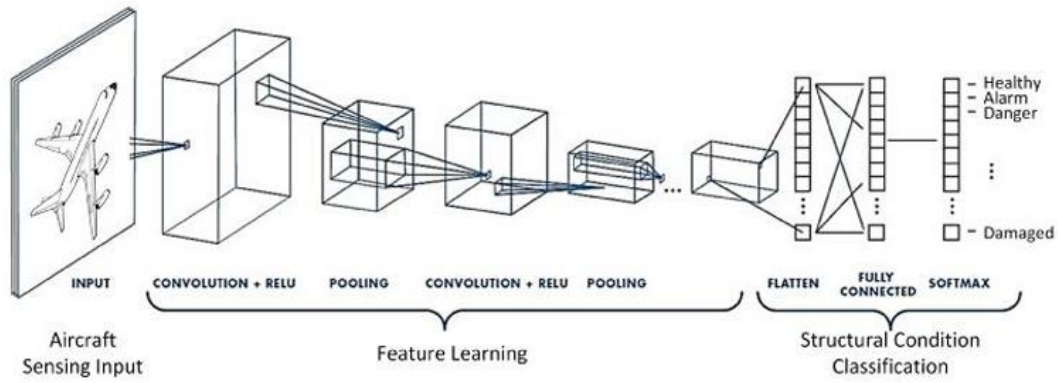


Figure 38: Convolutional Neural Network

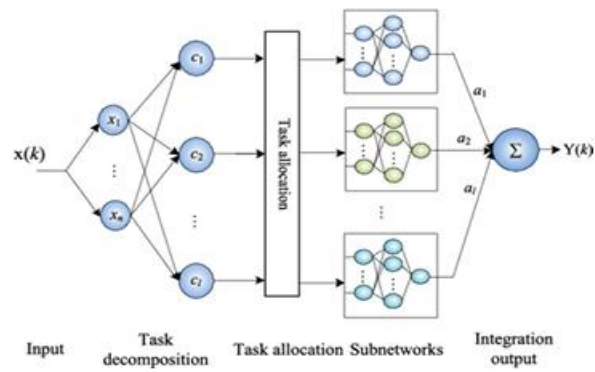


Figure 39: Modular Neural Network

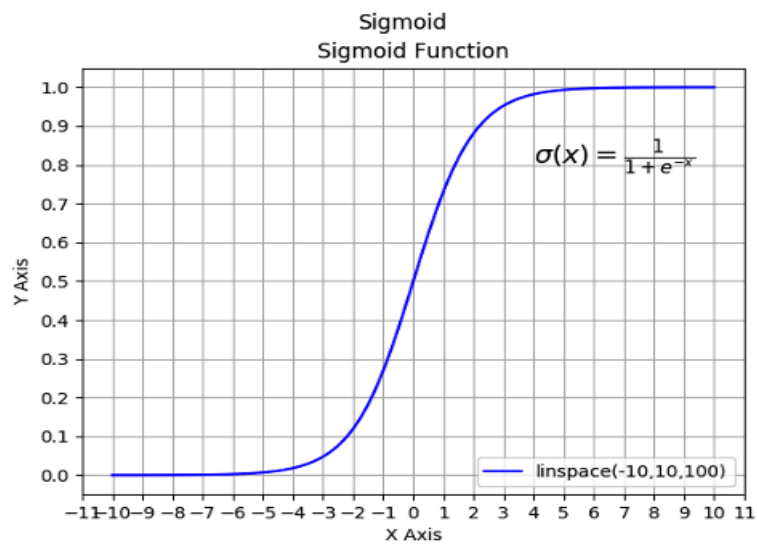


Figure 40: Sigmoid Function

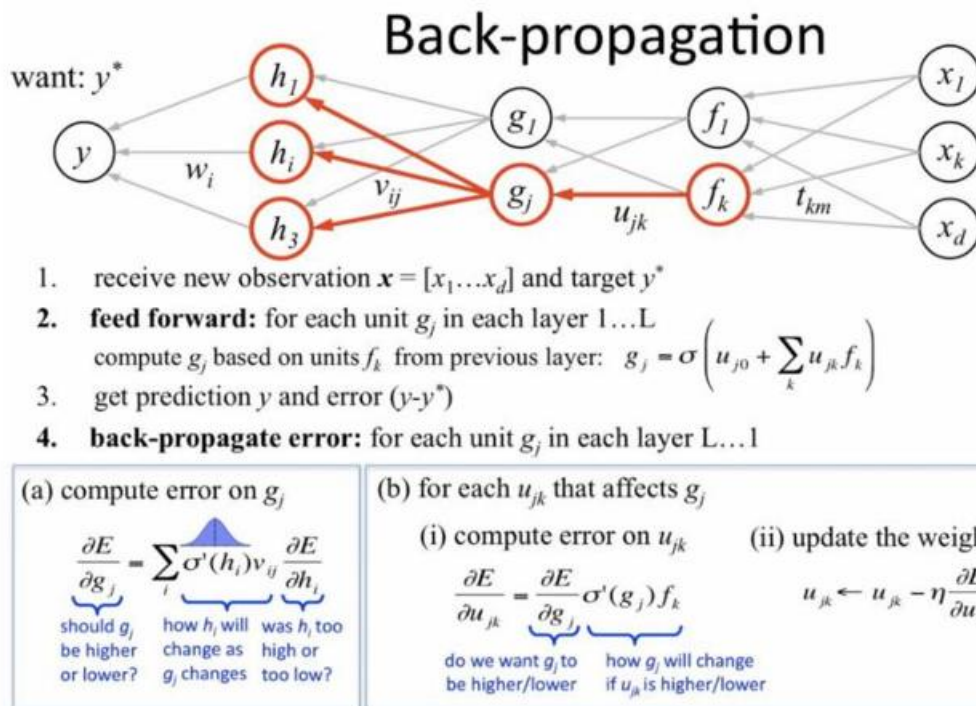


Figure 41: The Back Propagation Algorithm

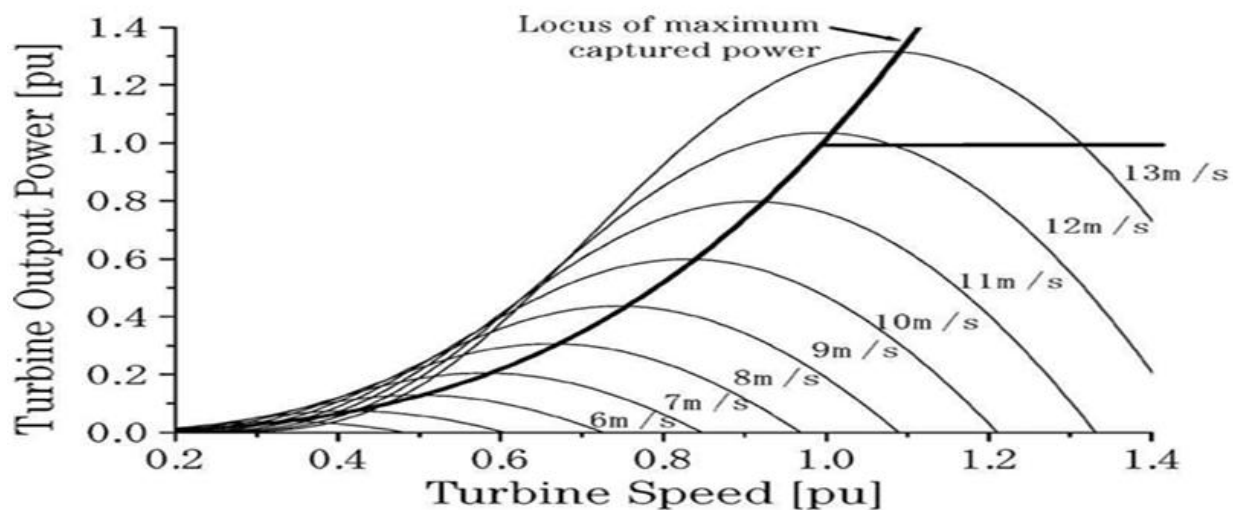


Figure 42: Wind turbine characteristics in terms of speed and power output

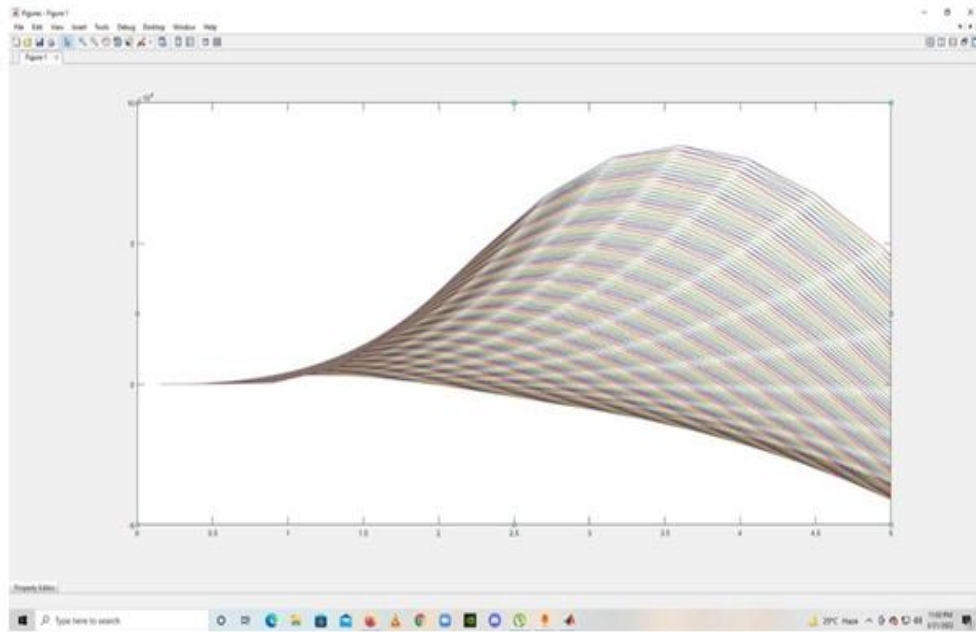


Figure 43: Characteristics curve of wind turbine

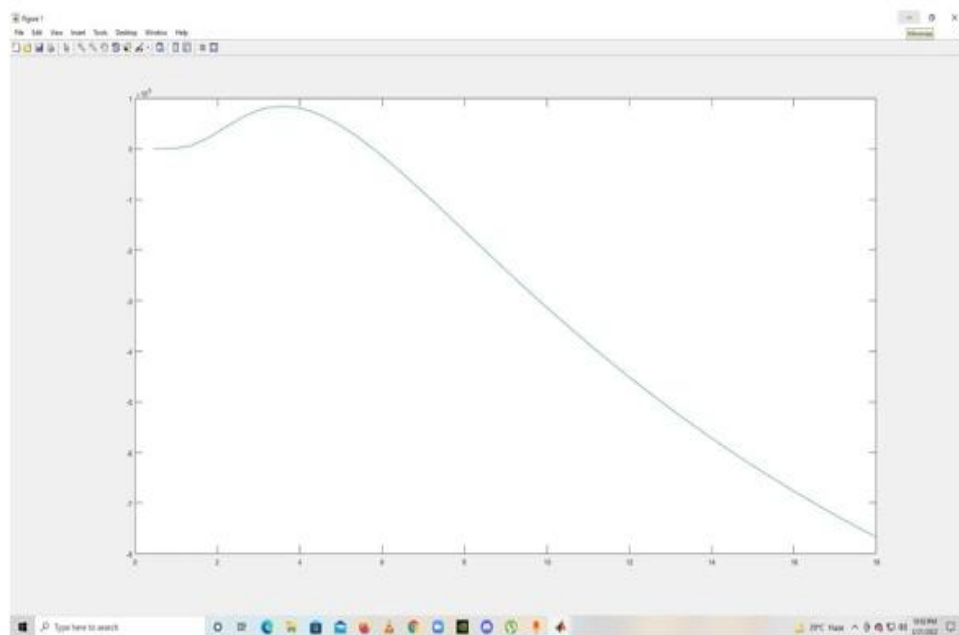


Figure 44: Characteristics curve of wind turbine(single)

D.

During the implementation phase, I created an ANN-based MPPT method in a step-by-step manner. Initially, I made the input and output profiles of the wind turbine characteristic curves. I then developed ANN-1 to predict wind speed with inputs of rotor speed, turbine power, and wind speed as the desired output. I thereafter used MATLAB nntool to design the network, choose the training procedure, and export the trained model. I then designed ANN-2 to predict optimal rotor speed using wind speed, and optimal rotor speed as output. I once again trained the network using nntool and then exported the trained system to be used eventually. Next, I linked both networks in series such that ANN-1 was the first network to estimate the wind speed, and ANN-2 estimated the optimum rotor speed to get the maximum amount of power. I monitored the performance of the regression and validation results after implementation. I also noticed that the results of training, validation, and testing were near one, indicating that the networks were effectively trained, and errors were minimal. This assignment included working on a realistic engineering issue, applying tools and models, developing a full process, operating the work systematically, and recording the technical result in a clear form.

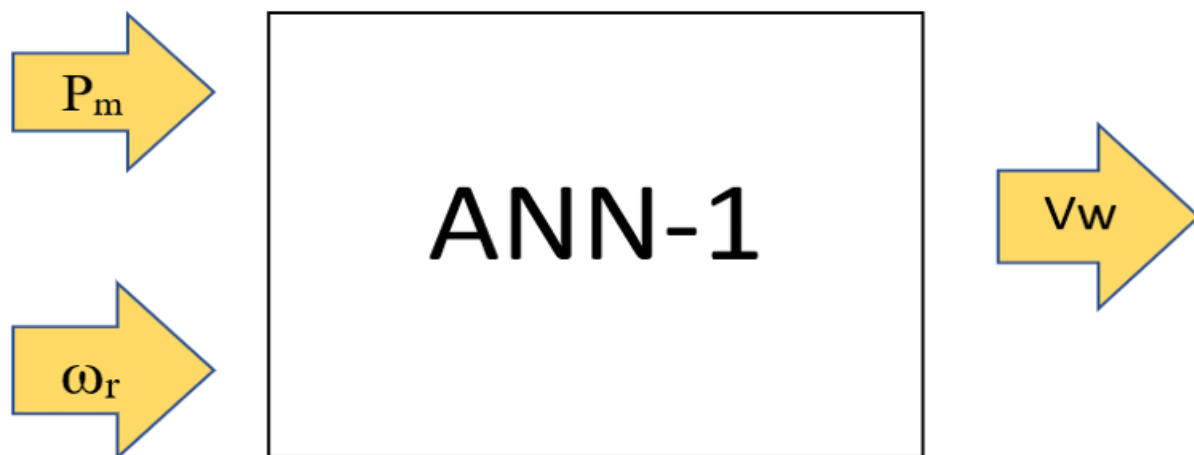


Figure 45: Wind Speed Estimation by ANN-1

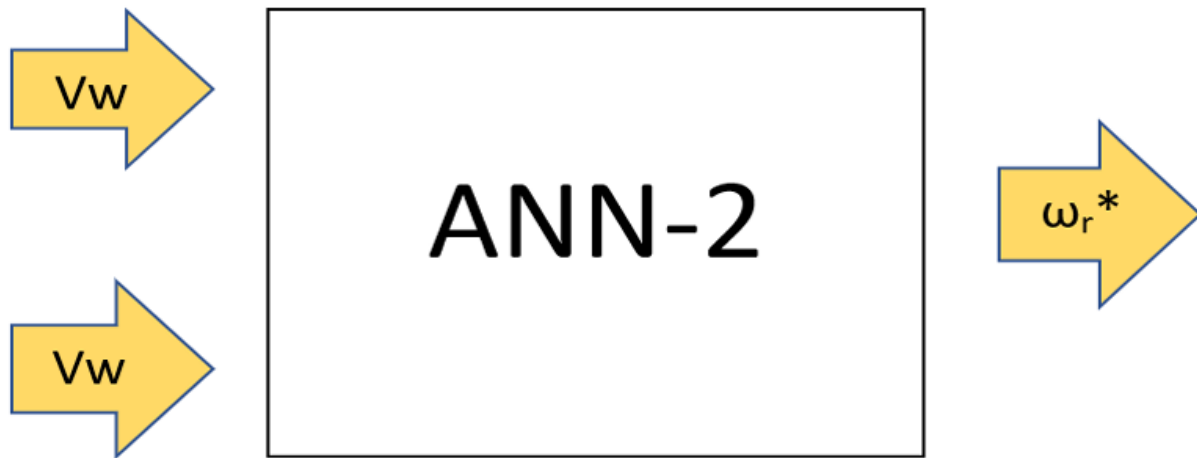


Figure 46: Optimal Rotor Speed Estimated by ANN-2

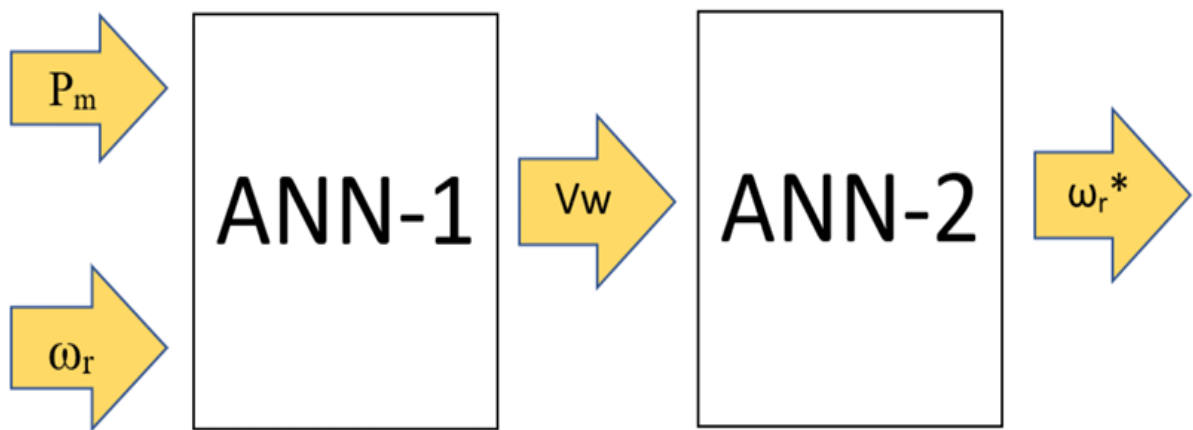


Figure 47: Proposed ANN based MPPT controller

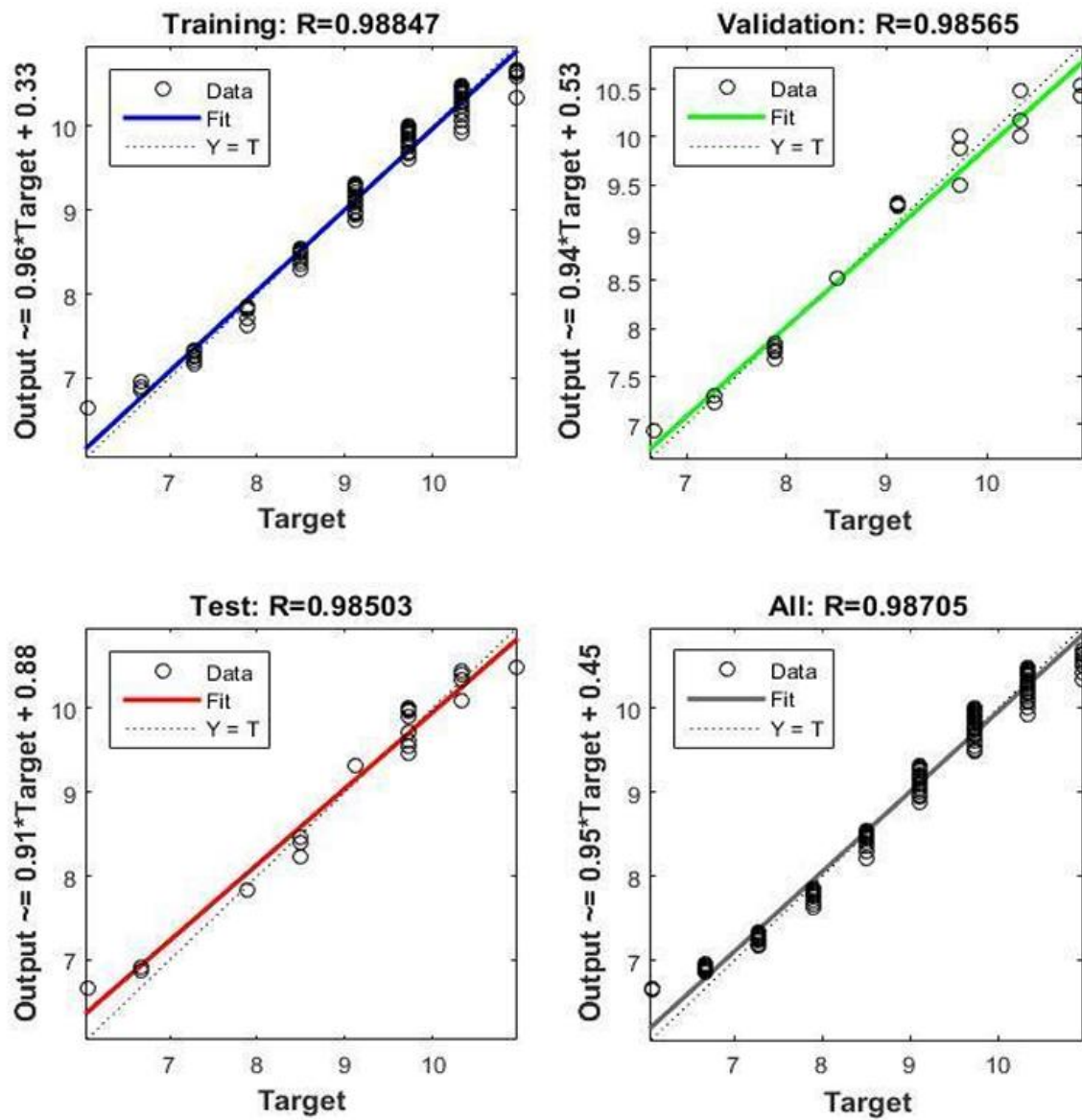


Figure 48: Regression Analysis of ANN-1

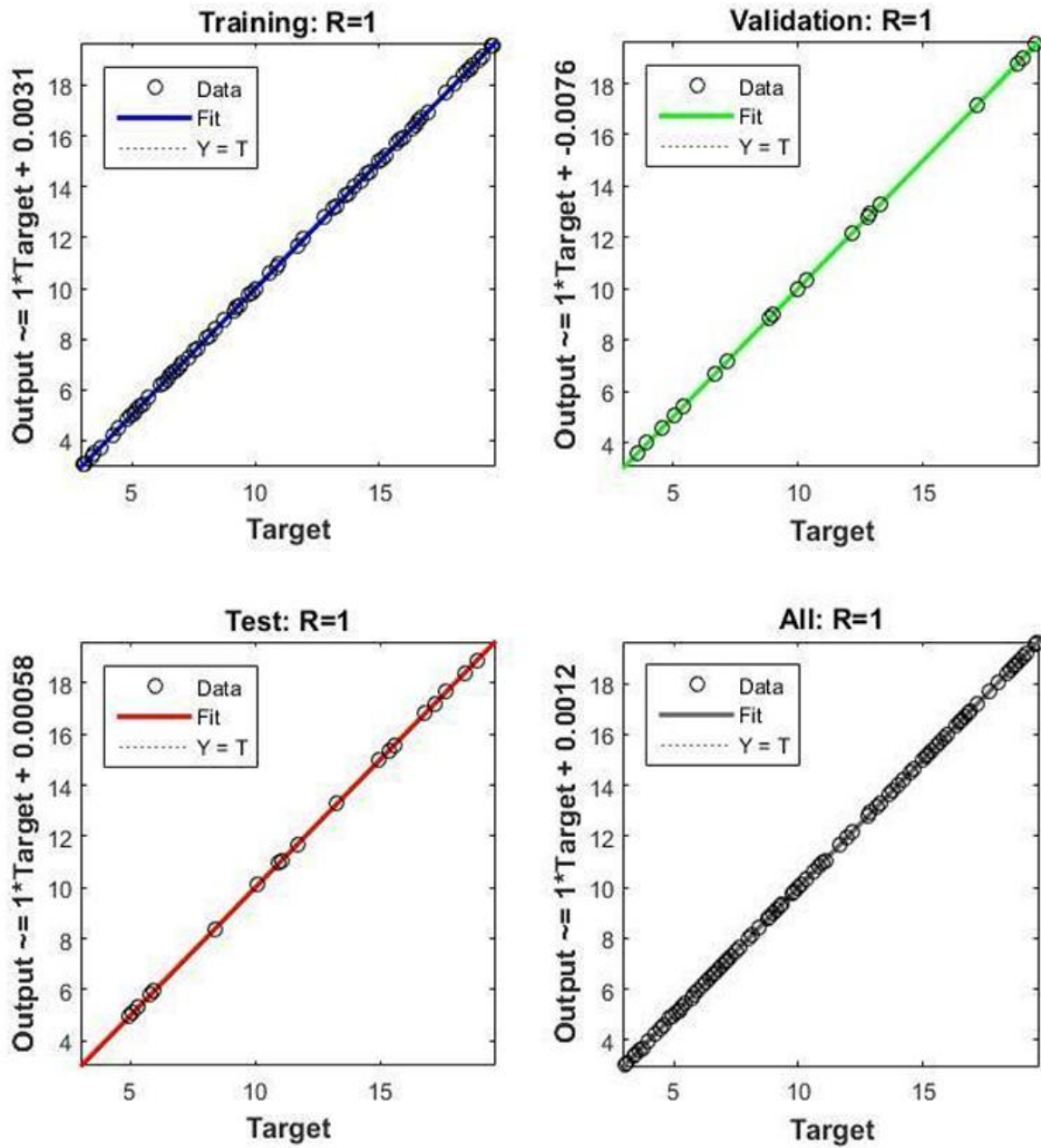


Figure 49: Regression Analysis of ANN-2

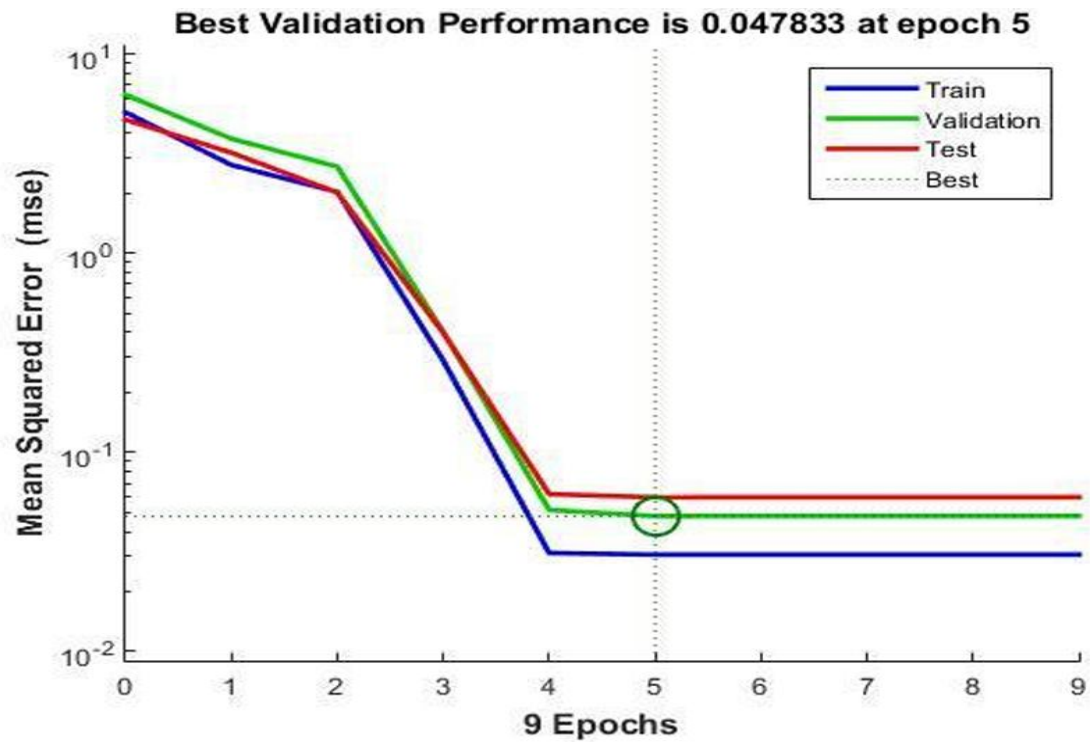


Figure 50: Validation Performance of ANN-1

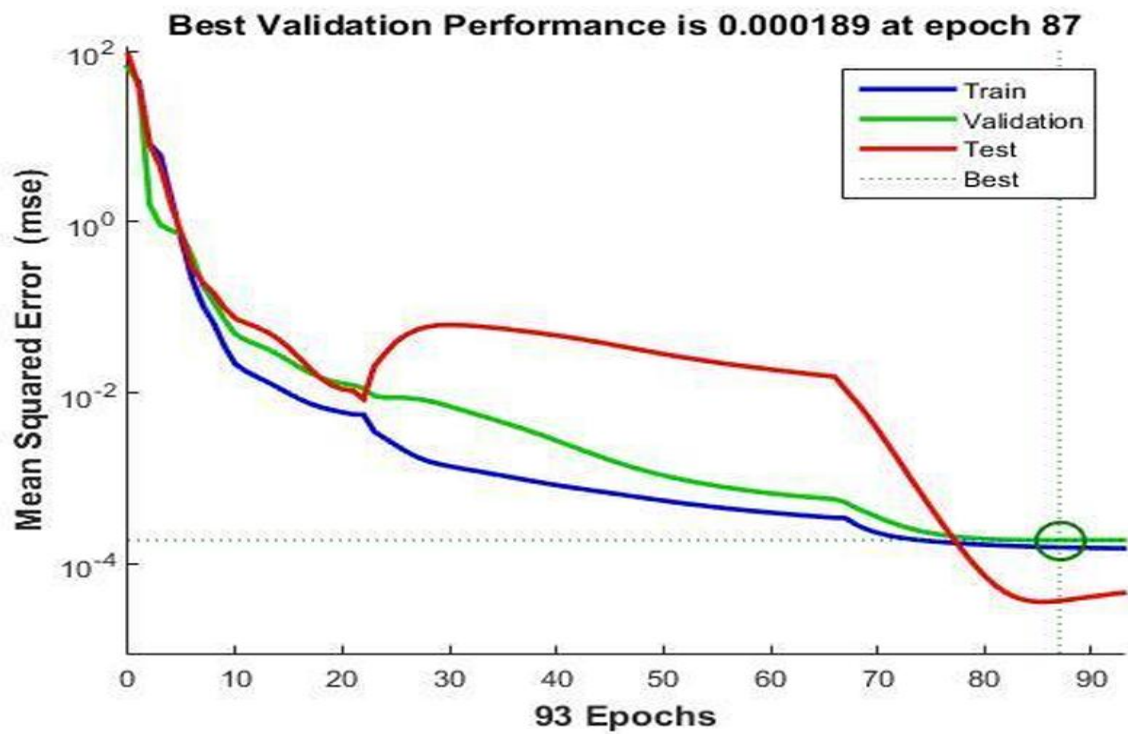


Figure 51: Validation Performance of ANN-2

E.

Lastly, I assessed the output of the ANN-based MPPT system in the case of speedily shifting wind conditions. I compared the initial wind velocity and the estimated one and observed that the two curves were nearly similar. Then I looked at the error of the wind speed estimation and found that the maximum error was 0.607m/s which was good in terms of estimation performance. Then, I also compared the original rotor speed to the estimated optimum rotor speed and discovered that the projected performance was closely related to the actual performance. I also verified the error in estimating the rotor speed and found that the maximum error was just 0.05332 rad/s. Based on these results, I concluded that the proposed controller was able to track maximum power and was effective without a mechanical wind speed sensor. Then I suggested further enhancement of the system, saying that only one ANN would be needed rather than two, which would allow making the system less expensive and easier. At this last phase, I showed data analysis, assessment of errors, technical judgment, sustainable thinking, and clear reporting. In such activities, I used responsible interpretation results, clear communication, organized work, and obtained a realistic conclusion in engineering.

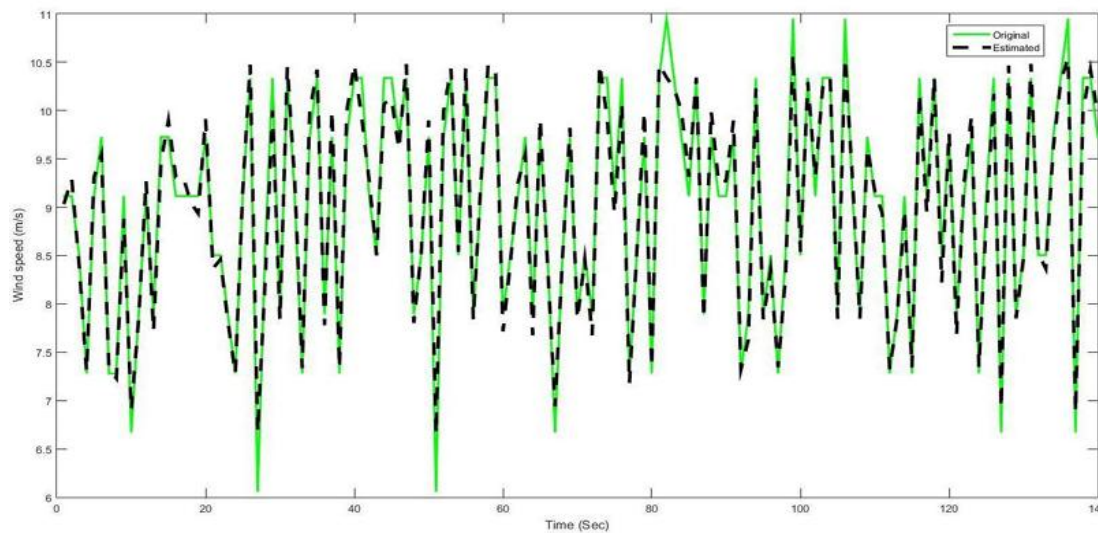


Figure 52: Original and Estimated Wind Speed

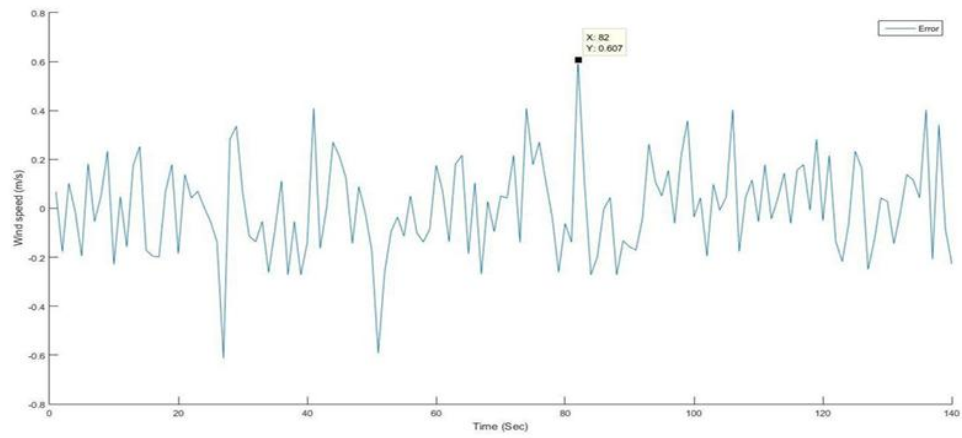


Figure 53: Wind Speed Error

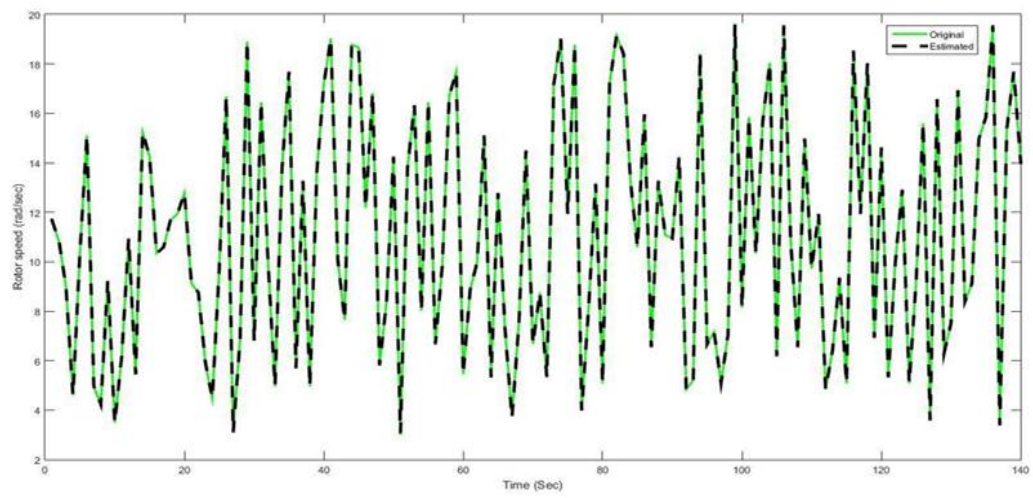


Figure 54: Original and Estimated Rotor speed

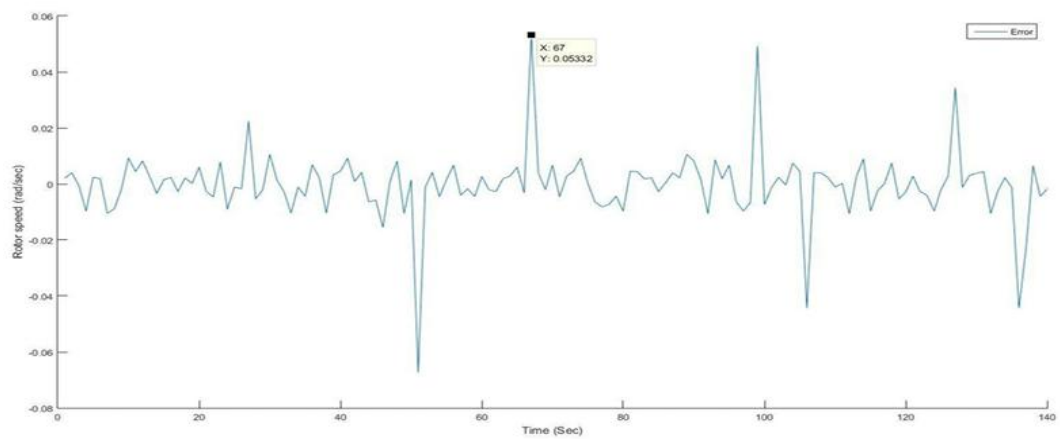


Figure 55: Rotor Speed Error

3.4 Technical Problems and Solutions

A.

I encountered an issue of poor estimation of wind speed without a mechanical sensor in an ANN-based MPPT system. This issue had an adverse impact on the system since the wrong estimation of the wind speed decreased the accuracy of maximum power extraction. I examined the cause and discovered the lack of diversity in training data and the restricted amount, which led to errors in estimation due to an inadequate ANN structure. I also noted that prediction became more difficult due to abrupt changes in wind. To address that, I enhanced the data by creating extensive wind turbine data in MATLAB using small step sizes and calculated input values such as rotor speed and turbine power very carefully. I trained the ANN appropriately with the help of appropriate algorithms. Once the implementation was done, I checked the results and realized that the error in estimation had significantly decreased, and performance was improved.

B.

Another issue that I encountered was finding the correct optimal rotor speed prediction to maximize power tracking. This issue had adverse impacts on system performance, given that the wrong rotor speed caused loss of power. I examined the root cause and discovered that when the ANN was improperly trained, and the connection between ANN-1 and ANN-2 was weak, thus decreasing scenario predictions. I also observed that system stability was influenced by variation in wind conditions. To address this, I have created two independent ANN models and related them together in sequence to enhance the accuracy of the prediction. I trained both sets of networks with correct validation and regression checks. Upon implementation, I assessed the outcomes and found that the rotor speed was predicted very well and the system was able to effectively track the optimal power output.

3.5 Creative Work

I refined the wind speed estimation by upgrading the dataset, training the ANN effectively, and minimizing errors. I also maximized the prediction of rotor speed with two distinct ANN models, which enhanced the accuracy and efficiency of the system.

3.6 Team Coordination

I led the project team by initially grouping work and delegating roles according to the strengths of the members. I organized meetings to talk about project objectives, project progress, and problems, so that everybody was aware of their role. I provided team guidance on data collection, MATLAB simulations, and the development of the ANN models, including support and clarification of any technical doubts. I

monitored the work regularly, checked the quality of outputs, and encouraged cooperation in solving the issues effectively. I also ensured proper documentation and reported on progress so that the team was on track. Whenever issues arose, I facilitated discussions to find solutions and maintained the schedule. By doing so, I managed to achieve a smooth cooperation, proper communication, and successful deployment of the ANN-based MPPT system, while maintaining the accuracy, efficiency, and learning.

3.7 Codes and Standards

I adhered to the standards of MATLAB code construction, coded datasets, created the network with the help of nntool, trained the network, applied the necessary checks of validation and regression, and described all of the work with the help of correct and well-organized data to obtain the ANN-based MPPT system.

3.8 Summary

A.

The project involved the enhancement of the maximum power recovery of a wind energy system through an ANN-based MPPT. It started with researching wind energy systems, finding out which MPPT techniques and components were appropriate in turbines, and learning how energy was converted and the environmental aspects of turbines. Artificial neural networks were then researched, and wind turbine data were organized to be modeled and trained accurately using ANNs. An ANN-based MPPT was designed and implemented in MATLAB to estimate wind speed and determine the optimal rotor speed for maximum power. Lastly, the performance of the system was evaluated through the analysis of errors and confirmed effective maximum power tracking under varying wind conditions. The program was successfully completed, and it demonstrated the existence of an effective sensor-less ANN-based MPPT system, which could possibly be further optimized with respect to its simplicity and costs.

B.

I enhanced my capability to handle wind energy systems and ANN-based MPPT technology in a good manner. I improved my communication, teamwork, and project management by planning and documenting in a systematic approach. I learned more and better about turbines, neural networks, and power optimization methods.